Calibrated Perplexity for Robust AI-Generated Text Detection

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Problem: Domain Shift in Perplexity

Same text, different perplexity across domains

Example: A news article vs a scientific paper

- News model: perplexity $= 11.7$
- Science model: perplexity $= 27.3$

Single threshold fails across genres.

Research question: How to make perplexity-based detection robust to domain shift?

Related Work

- GLTR: Statistical detection using top-k rank analysis
- Binoculars: Zero-shot detection using cross-perplexity
- Chen et al.: Smoothing techniques for n-gram LMs
- Bakhteev et al.: Domain shift and length impact
- Gritsay et al.: Fragment detection in academic texts

No systematic approach to calibrate perplexity across different domains.

Problem Statement

Key observation: For the same text x, perplexity varies across domains:

$$\mathrm{PPL}_{\mathcal{M}_{\text{news}}}(x) \neq \mathrm{PPL}_{\mathcal{M}_{\text{science}}}(x)$$

Our goal: Make perplexity domain-invariant through calibration.

Minimization problem:

$$\sum_{i=1}^{N_d} \sum_{x_j \in X_i} (M_{\mu_i,i}(x) - y_{ij})^2 + \sum_{i<j\leq N_d} (\mathrm{TH}_{M_{\mu_i,i}}(x) - \mathrm{TH}_{M_{\mu_j,j}}(x))^2 \to \min_{\bar{\mu}}$$

We minimize the variance of thresholds in ensemble across different domains for the same text x, making the score domain-invariant.

Our Solution: Ensemble of Domain-Specific Models

Key Idea:

1. Train separate KenLM models on each domain
2. For a new text, compute perplexities from all models
3. Combine predictions via weighted voting

Advantages:

- No need to know the test text's domain
- Each model specializes in its domain
- Weighted voting balances human/AI detection

Methodology

Training:

- Train trigram KenLM models on human texts from 7 domains
- Domains: news, hard_essays, scientific texts, stories, articles, hard_news, hard_science

Threshold Calibration:

$$\tau_d = \arg\max_{\tau} \mathrm{F1}\left(\mathbf{1}[\mathrm{PPL}_{\mathcal{M}_d}(x) < \tau],\ y(x)\right)$$

Ensemble Decision:

$$H(x) = \mathbf{1}\left[\sum_{d \in \mathcal{D}} w_d \cdot \mathbf{1}[p_d < \tau_d] \geq 0.5\right], \quad w_d = \frac{\mathrm{BalancedAcc}(\mathcal{M}_d)}{\sum_{d'} \mathrm{BalancedAcc}(\mathcal{M}_{d'})}$$

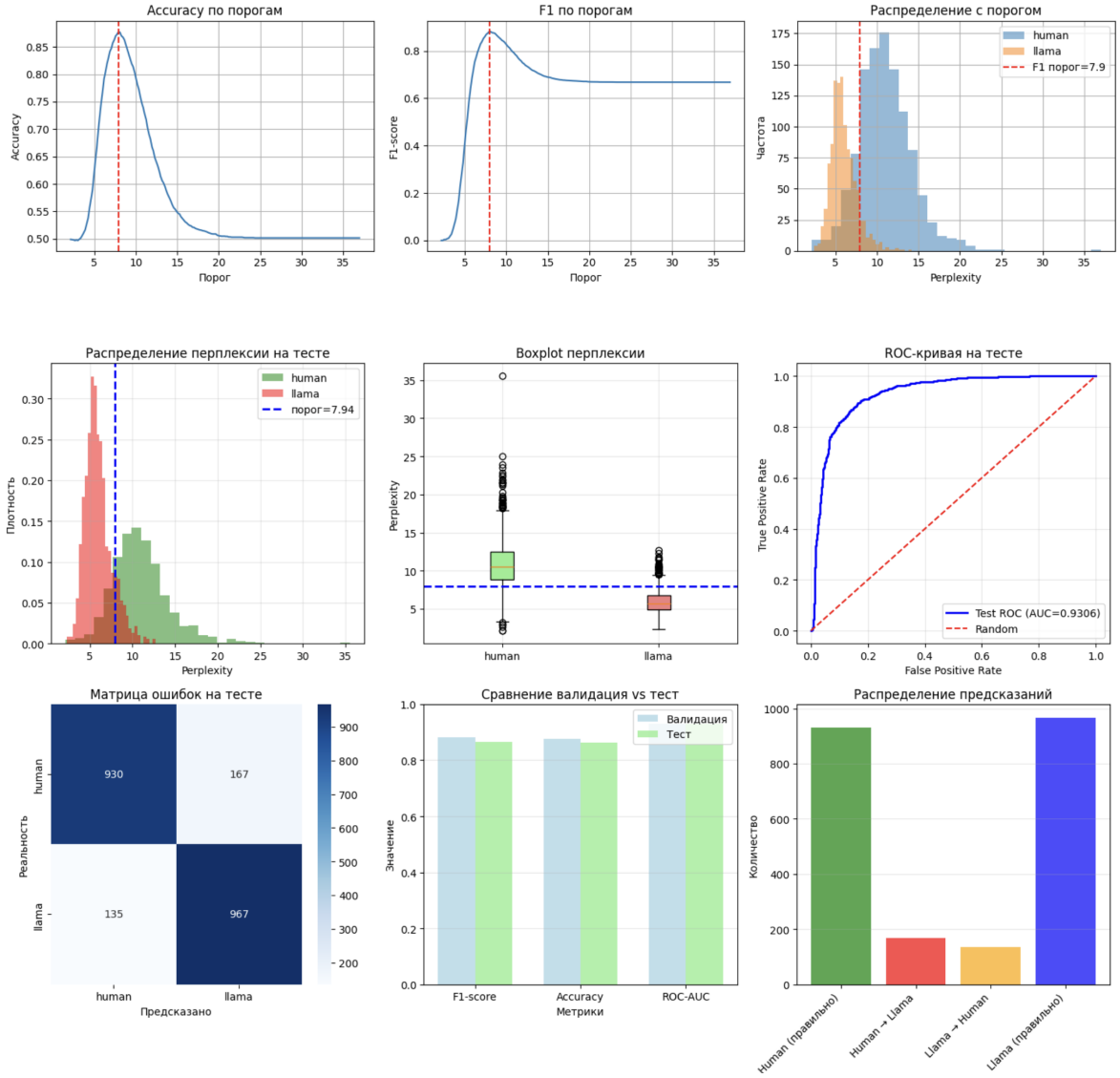
Experiment 1: Cross-Domain Perplexity Matrix

Train/Test	AINL	stories	reviews	articles	
AINL (science)	**11.6 / 6.9**	27.3 / 23.4	26.1 / 21.9	23.8 / 19.2	22.
stories	26.8 / 22.7	**16.8 / 10.6**	16.1 / 14.2	20.0 / 18.7	18.
reviews	23.3 / 18.0	18.5 / 13.8	**11.1 / 9.3**	18.8 / 15.8	16.
articles	19.9 / 14.2	20.1 / 15.2	17.7 / 13.6	**13.3 / 9.0**	17.
news	22.0 / 16.8	20.0 / 15.3	18.1 / 15.1	19.2 / 16.3	**11**
HARD_science	15.7 / 11.2	21.9 / 18.0	21.6 / 17.2	20.0 / 16.9	18.
HARD_essays	20.4 / 17.2	18.8 / 15.3	19.8 / 16.3	19.7 / 18.5	19.
HARD_news	18.4 / 15.5	18.1 / 15.5	18.0 / 15.4	17.4 / 16.2	14.

Cell format: Human PPL / AI PPL. **Bold diagonal** = in-domain performance.

Observation: Perplexity increases significantly when moving off-diagonal, confirming domain shift.

Additional Visualizations



Experiment 2: Tokenization and Pruning

Goal: Improve model stability across domains

Methods tested:

- SentencePiece tokenization (vocab_size=8000)
- N-gram pruning (–prune 2 2 2)
- Size cutting
- Combined optimization

Result: Only 1–2% improvement – helpful but not sufficient.

Experiment 3: Ensemble Strategies

We compared four ensemble strategies:

Results:

Method	Human recall	AI recall
Simple majority	32%	91%
Weighted (val acc)	78%	58%
Weighted (balanced acc)	73%	66%
Perplexity-weighted	93%	33%

Key insight:

- Simple majority excels at machine detection (91%)
- Perplexity-weighted excels at human detection (93%)
- Weighted by balanced acc gives best trade-off

Proposed combination: Use difference between perplexity-weighted and simple majority to achieve high quality on both classes.

Key Results for now

Best-performing method: Weighted voting by balanced accuracy

Metrics:

- Balanced Accuracy: 0.69
- F1-score: 0.68
- Human recall: 73%
- AI recall: 66%

Comparison: Most balanced human/AI detection (gap reduced from 58% to 7%).

Conclusion

Key Contributions

- Demonstrated that perplexity thresholds are highly domain-dependent (vary by 2–3x)
- Proposed ensemble of domain-specific KenLM models for domain-agnostic detection
- Showed that weighted voting by balanced accuracy achieves the most robust results
- Achieved balanced accuracy of 0.69 with human/AI recall rates of 73%/66%

Future Work

- Upgrade combined ensemble
- Test on more diverse domains and languages
- Combine with neural-based detectors